

CS 747 Foundations of Intelligent and Learning Agents

Assignment 2

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1 Task 1

1.1 Policy Value Evaluation

- In the algorithm implemented, the function *val_policy* evaluates the value function for a given policy $\pi(s)$
- The value function is obtained by using the Bellman equation

$$V^\pi(s) = \sum_{s' \in S} T(s, \pi(s), s') \{R(s, \pi(s), s') + \gamma V^\pi(s')\} \quad (1)$$

- We define two value lists, *value*, *new_value* and when the error between these is less than 10^{-12} then the loop ends and gives the final value function

1.2 Value Iteration

- We have two lists here as well, *value* and *new_value*. *Value* is initially defined as random
- We then iterate and find the *new_value* using the value iteration below

$$V_{t+1}(s) = \max_{a \in A} \sum_{s' \in S} T(s, \pi(s), s') \{R(s, \pi(s), s') + \gamma V_t(s')\} \quad (2)$$

- We continue until the precision among all the elements is $1E-10$ and break the update loop
- Finally we find the policy corresponding to this value function by iterating over all actions and finding the policy with the value function obtained above

1.3 Howard's Policy Iteration

- We start with a random policy named *policy* and another policy *new_policy*
- We then find improvable actions corresponding to each state, and if some action *a* is better than that stored in current policy (its action value function) it's a improvable state

$$Q^\pi(s, a) = \sum_{s' \in S} T(s, a, s') \{R(s, a, s') + \gamma V^\pi(s')\} \quad (3)$$

- We than update this in the new policy and iterate until the new policy is same as the old policy for all states
- Here, we break the loop and then return the value function (corresponding to the latest updated policy) and the policy

1.4 Linear Programming

- Here, we directly use the pulp linear programmer, we give input to it to maximize the constraints

$$\text{maximize}(-\sum_{s' \in S} V(s)) \quad (4)$$

subject to the condition

$$V(s) \geq \sum_{s' \in S} T(s, a, s') \{R(s, a, s') + \gamma V(s')\} \quad (5)$$

- This gives us the value function and then we use this to find the corresponding policy
- This linear program has n variables and nk constraints and gives us the value function

2 Task 2

2.1 MDP Formulation

- The mdp made has $16 \times 16 \times 16 \times 2 + 2$ states. Each 16 corresponding to the 16 positions of the players B1, B2, R and 2 corresponding to the possession (either B1 or B2). The last 2 states correspond to goal or end state
- The states are numbered from 0 to 8193, where 8192 corresponds to goal and 8193 corresponds to end state. Other correspond to different locations patterns of B1, B2 and R
- The discount factor used is 1, as it is a episodic task
- Functions are made to find x and y co-ordinate for each 1 to 16, function to give state corresponding to the state given in the opponent policy file
- Different functions are made to shoot, move and passing. Functions such as inline are made to check if B1, R, B2 are in line either on x or y or x-y, x+y as explained in the moodle announcement
- In shoot, if R lies on location 8 or 12, then the probability is halved i.e. it becomes

$$\text{prob_values}[i] \cdot 0.5 \cdot (q - 0.2 \cdot (3 - x1))$$

where prob_values[i] denote probability of R moving into locations 8,12 by whatever L, R, U or D. The location is first checked and the probability is checked

- If R does not go to 8 or 12 then probability is not halved and is equal to

$$\text{prob_values}[i] \cdot (q - 0.2 \cdot (3 - x1))$$

- For end state, the probability of transition is $1 - \text{prob_sum}$ where prob_sum corresponds to each 4 movements of R so that it goes to different states

- For passing, if B1, R and B2 lie in a straight line as mentioned above (point 5) the probability is halved i.e. it becomes

$$\text{prob_values}[i] \cdot 0.5 \cdot (q - 0.1 \cdot \max(|x1 - x2|, |y1 - y2|))$$

if it does not lie in a straight line then it will be

$$\text{prob_values}[i] \cdot (q - 0.1 \cdot \max(|x1 - x2|, |y1 - y2|))$$

- For going to the end state, it will be $1 - \text{prob_sum}$ where prob_sum is the sum of probabilities of R going to any of the other locations
- In the passing function, tackle is also considered initially as well as when B1 goes out of the 16 block and B2 goes out of the 16 blocks.
- There are 4 conditions, when possession is with 1 and 1 moves, 2 moves and vice versa conditions
- In the first condition when possession is with 1 and 2 moves, probability of moving in correct direction is $(1 - p) \cdot \text{prob_values}[k]$ rest $1 - \text{prob_sum}$ is of going to the end state
- When possession is with 1 and 1 moves, then we check for tackling first, if tackle happens, that is final position of both B1 and R is same of they swap places, then probability is halved and becomes $0.5 \cdot (1 - 2p) \cdot \text{prob_values}[k]$ and probability of ending is $((1 - 2p) \cdot 0.5 + 2p) \cdot \text{prob_values}[k]$ and if tackling does not happen then probability of ending is $2p \cdot \text{prob_values}[k]$ and probability of pass is $(1 - 2p) \cdot \text{prob_values}[k]$
- Same for the other 2 cases
- In the decoder, we are using the opponent file and our value file and copying the states from the opponent to our corresponding states 1 to 8192 and replacing them with the original states from opponent file
- The value of gamma, i.e. discount is kept 1 because we want to find the estimate number of goals.
- The reward is one only when there is a goal, i.e. goal on shooting else it is always 0.

2.2 Graphs and Explanation

2.2.1 Graph 1

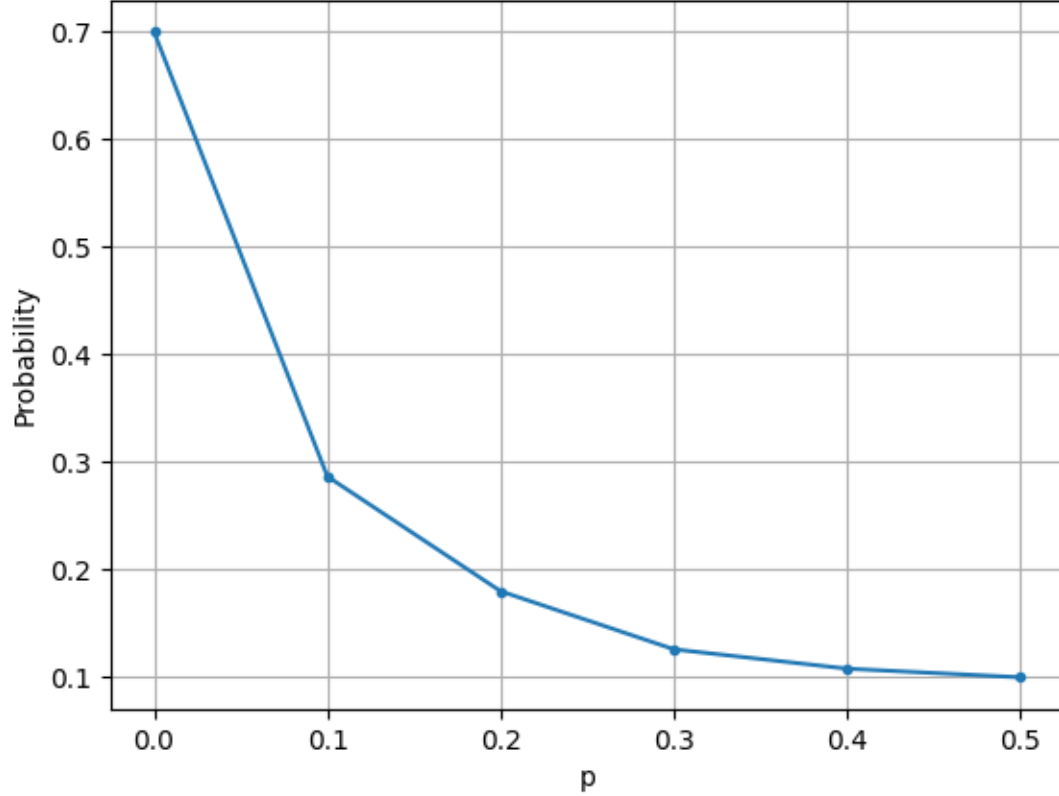


Figure 1: Probability of Winning vs p , $q = 0.7$

2.2.2 Explanation for Graph 1

The probability of movement in desired direction is $1-2p$, $2p$ is probability of losing possession directly, p tells how poor the player is for movement. Higher value of p means that the player is not able to move properly also higher value of p shows higher probability of tackling. Hence as p increases, the expected probability of goals scoring and also winning reduces. This is intuitive. Also the policy is greedy defense, moving towards the player with possession, hence R moves towards the player with possession, which means that it is highly to come in the direction of either between $B1$ and $B2$ and reduce passing probability (may come in between them), also it may come between the shooter and the goal so shoot probability also reduces. This explains that as p will increase, the probability of goals, and the expected probability of winning will decrease. Also, q is kept constant at 0.7, so shoot and passing probability remain same. The intuition is well explained by the resulting plot of decreasing probability of winning as p increases. We observe that probability of winning is 0.7 at $p=0$ and reduces to 0.1 as p reaches 0.5. Hence, we observe that as p increase the winning probability decreases.

2.2.3 Graph 2

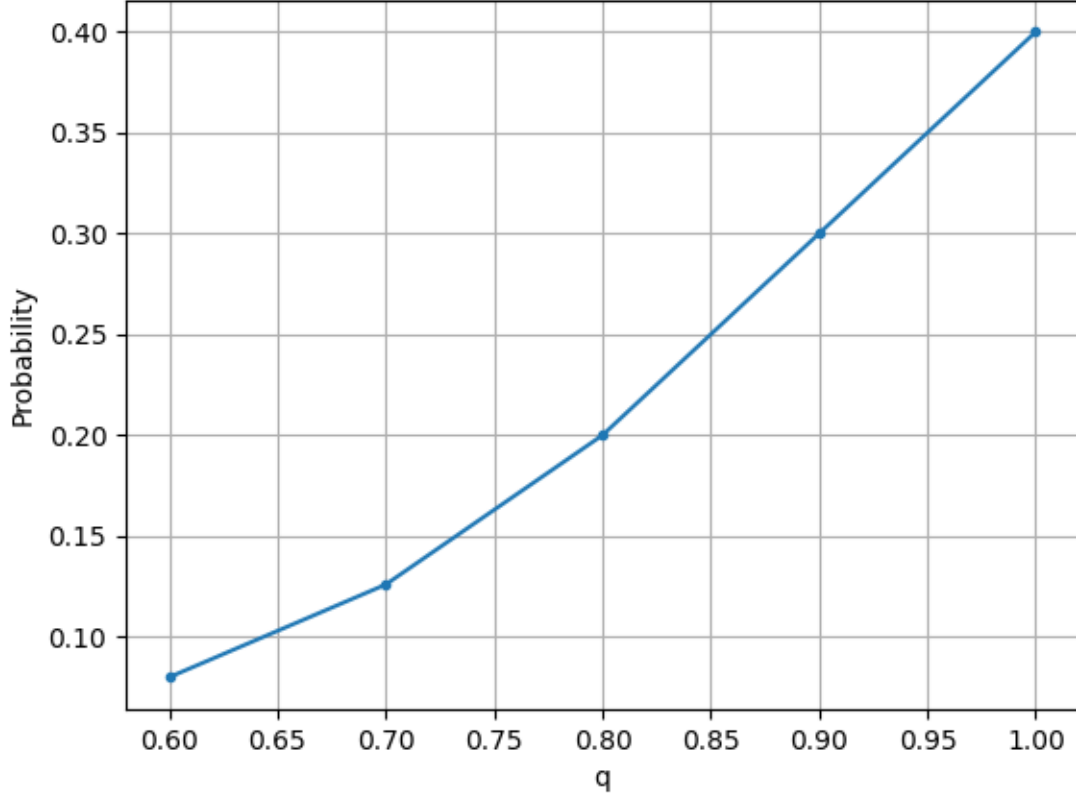


Figure 2: Probability of Winning vs q , $p = 0.3$

2.2.4 Explanation for Graph 2

Here, p is kept constant so movement probability is constant. As the probability of q is being increased, this means that probability of successful passing and shooting are increasing. Also greedy algorithm is being used by the opponent, so whenever the opponent approaches a player with possession supposed B1 without loss of generality, B1 can pass to B2 and B2 can shoot, this would result in higher winning probability. Also, as p is increasing the probability of successful pass and shoot is also increasing so passing can be done more successfully and shooting as well. So if a player R approaches B1 it can either shoot or pass with higher probability now. Hence the number of goals scored and hence probability of winning increases as q increases. We observe that probability of winning is 0.05 at $q = 0.6$ and increases to 0.4 when q reaches 1 (p is kept constant). This is observed in the plot given as well. Hence our intuition is well explained or validated.